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TANK FOUNDATIONS

PURPOSE

This document establishes guidelines and recommended procedures for the design of tank foundations.

SCOPE

This document provides the following:

- Discussion on the various types of tank foundations.
- Special conditions which must be considered in foundation type selection and design.
- A detailed procedure for concrete ring wall design.
- Two sample designs for concrete ring wall foundation.

APPLICATION

The types of storage tanks normally encountered in hydrocarbon, petrochemical, and other industrial plants have cylindrical shells, essentially flat bottoms, and either cone roofs or floating roofs. Tank size may range from 10 to over 200 feet in diameter, with heights from 16 to 56 feet. Due to the wide variety of surface, subsurface, and climatic conditions, it is impossible to specify one type of foundation or design procedure for all storage tanks. Each case should be analyzed individually and a foundation designed to meet those individual requirements.

TYPES OF FOUNDATIONS

The types of foundations most commonly considered include the following:

- Gravel ring walls supporting the tank shell.
- Concrete ring walls supporting the tank shell.
- Concrete mat on grade supporting the entire tank.
- Concrete mat on piles supporting the entire tank.
- Concrete rib beams and mat foundation for cold storage tanks.
- Concrete mat and heating coils for cold storage tanks.

The type of the foundation depends on design requirements, type of tank, tank dimensions, soil and site conditions, environmental conditions, material availability, local codes, and Client requirements. A number of other conditions, which may have a significant impact in the foundation design, need to be considered for selection of the type of foundation. They include the following:

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- Corrosion
- Temperature fluctuations
- Frost
- Proximity to slope
- Use of sea water for compaction

A brief discussion of the types of foundations and special conditions to be considered is presented in the following sections.

GRANULAR RING WALL FOUNDATIONS

This is the most economical and common type of foundation used for storage tanks. This type of tank foundation involves a ring wall of coarse granular material supporting the tank shell with the tank bottom resting on compacted fill overlain by asphalt sand. Refer to Civil Engineering Standard Drawings for typical details for storage tank foundations.

The foundation is used in the following:

- Tank farm areas with tank diameters of about 12 feet or larger.
- No internal pressure, or other need to bolt tank down.
- Tank sites where subsurface conditions and settlements are within acceptable limits.
- Areas where there is good availability of granular material.
- Regions of low seismicity or in seismic areas where the tank diameter to height ratio is such that there is no uplift of the tank shell or small sloshing effect. The magnitude of lateral forces, overturning moments, and associated hydrodynamic mass must be determined to assess their impact on tank shell and foundation design.

Leveling Ring

A concrete leveling ring can be used under the tank shell in the gravel ring wall foundation. This leveling ring is 8 inches deep by 12 inches wide unreinforced concrete whose primary function is to provide a stable level base upon which the fabricator can build the tank shell wall. Additional advantages of the leveling ring are to distribute concentrated shell loads on to the gravel ring wall and minimize edge settlement under seismic condition.

Fuel Oil Sand versus Asphalt Sand

The main reasons for using asphalt sand instead of fuel oil sand are environmental and safety. In many areas, local regulations do not allow use of fuel oil for spraying on slopes for slope stabilization or mixing with soil to prepare an oil sand base for tank bottoms.

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Other Considerations

Special treatments due to project or environmental requirements.

CONCRETE RING WALL FOUNDATIONS

This foundation is primarily used in areas or situations where the gravel ring wall is not feasible. Refer to Civil Engineering Standard Drawings for typical details of storage tank foundations. This would include the following:

- In congested areas with space limitations.
- In seismic areas where tank shell must be anchored to the foundation to provide uplift resistance.
- In areas with marginal soil conditions where re-leveling may be necessary due to differential settlements.
- To bridge over localized soft spots.
- For pressurized tanks.
- For floating roof tanks.
- To extend the foundation below frost lines.
- To accommodate Client requirements or preferences.

CONCRETE MAT FOUNDATIONS ON GRADE

This foundation type is used primarily:

- For small diameter tanks (12 feet or less) due to ease of construction.
- To provide overturning seismic resistance where ring wall design is unable to withstand uplift forces in high seismic areas.
- For cold storage tanks.
- When large differential settlements are anticipated.
- For pressurized tanks.

CONCRETE MAT FOUNDATIONS ON PILES

This foundation type is used in very poor soil conditions where either the allowable soil bearing capacity is not adequate or the estimated settlements are more than the tank can tolerate. Surcharging is commonly used in poor soil areas to improve the soils for supporting tanks, but in many cases, sufficient time or the surcharge fill are not available.

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In certain instances, settlement tolerances are very small due to the sensitive nature of the stored materials or tank design which may require the tank to be supported on piles.

This type of foundation is generally not feasible for tanks over about 25 feet in diameter.

SPECIAL CONDITIONS

Corrosion

Corrosion of the tank bottom plates and how to prevent it must be considered in the design of the foundation system for storage tanks. There are several causes which can initiate and accelerate corrosion of tank bottoms. Three of the primary causes are as follows:

- The principal cause is nonuniform support of the tank bottom. This may be the result of warping of the bottom plates due to welding during tank erection. Or it may be due to point loading caused by foreign matter under the tank bottom such as wood pieces, stones, or excess sprayed asphalt that causes a lump on the surface of the support pad. This nonuniform support with point contacts provides a starting point for corrosion to develop under the tank. The presence of contaminants such as salts and moisture will accelerate this process.

- A second cause of corrosion is alternate wetting and drying in the presence of oxygen and heat. This is generally the case for fuel tanks where temperature cycling of the stored oil causes the tank base to breathe. A porous base contains ample water vapor and oxygen and can cause great volumes of these gases to move back and forth beneath the bottom plate. The tank bottom will be cooler at the edge than at the center, and water vapor moving outward will condense near the edge. As the tank cools, inflow of air will dry the steel. This alternate wetting and drying in the presence of oxygen and heat provides an ideal condition for corrosion to occur. It is clear that reduction in oxygen and moisture under the tank will minimize the corrosion potential. Therefore, supporting the tank bottom on an oiled sand or asphalt sand will minimize the supply of vapor and oxygen while protecting the plate with a film of oil or asphalt.

- Another case that could cause corrosion of tank bottoms is stray currents at the site emanating from neighboring cathodically protected underground pipes and structures.

Preventative measures include the following:

- Support pad should be constructed with salt free sand and be free of any foreign matter and lumps.
- Support pad should be protected from airborne or surrounding area contamination (some contaminants are brought to pad by workers shoes, such as clay and lumps).

- Minimize the time that tank will remain empty on its foundation after erection. In large projects, where construction schedules are long, effort should be made to at least partially fill the tanks in order to seek full contact between the bottom and the pad.
- Where cathodic protection is adopted for the protection of tank bottoms, the system should be energized as soon after completion of tank erection as possible.
- Investigate neighboring cathodic protection systems for presence of stray currents at the site.
- Seal the foundation by using oil sand or asphalt sand layer under the plate to prevent moisture and oxygen penetration under the bottom of the tank.

Frost

Three requirements must be satisfied simultaneously for frost heave to be a problem:

- Soil moisture supply.
- Cold temperature to cause soil freezing.
- Frost susceptible soils.

Two processes are involved in frost heave:

- Freezing of in situ pore water causes a 9 percent volume change of the water. This does not depend on the frost susceptibility of the soils. Assuming 100 percent saturation of compacted gravel, the anticipated heave may be on the order of 2 percent of the thickness of the soils affected by freezing. If the degree of saturation is less, the heave may not occur at all.

- Ice lensing by water migration into frost susceptible soils. This heave may be as much as 10 times greater than the heave caused by freezing of in situ moisture. This is the portion of the heave which is eliminated by using non-frost susceptible soils and is the target of measures adopted to reduce frost heave.

The depth of frost penetration or thaw (for permafrost areas) depends on the air temperature and the thermal conductivity of the materials. Gravels and crushed rock, the commonly used ring wall and nonfrost susceptible materials, would generally be subjected to increased depth of frost penetration than clayey or silty soils with vegetative cover. In selecting the thickness of in situ materials to be removed and replaced with nonfrost susceptible soils, this factor should be considered. The thickness of material replaced should be sufficient to prevent frost penetration through the replaced material into underlying frost susceptible materials. This thickness may be greater than the depth of frost penetration into natural silty or clayey soils.

The draining of the granular nonfrost susceptible material, although desirable, may not be necessary considering the cost involved and its potential impact on frost heave. Special measures for drainage (other than normal drainage of the area) may not be warranted by the frost heave considerations. Refer to Civil Engineering Standard Drawings for typical details of storage tank foundations for areas where frost heave is a problem.

- The total thickness of granular fill ***(H + h1)*** should be selected so that frost penetration through this material does not reach the frost susceptible underlying soils. This thickness may be greater than ***D***, the depth of frost penetration in natural surrounding soils. ***(H + h1)*** should be selected on the basis of maximum recorded depth of frost penetration in similar (gravel or crushed rock) materials in the area.

where

H = depth of nonfrost susceptible material below the bottom of the ring wall.

h1 = height of the ring wall.

- If *H* is large or suitable nonfrost susceptible materials are not available, a layer of insulation may be placed in the fill to reduce the thickness. The insulating effects of the tank and the product should also be considered.

- If penetration and mixing of the granular material with underlying silt is a potential problem, a visqueen or asphalt lining should be considered.

- The extent to which the frost susceptible materials are to be replaced under the tank depends upon the differential settlement or heave that can be tolerated. Tanks can generally withstand fairly high uniform settlements on the order of several feet. Differential settlement, or heave, near the shell should be limited to d/8 to d/13 where d is the distance over which the settlement, or the heave, occurs or as specified by the tank design, whichever is less. Using the maximum anticipated heave of h_a in the frost susceptible areas toward the middle, the minimum distance from the shell to which the frost susceptible soils should be replaced is about 13 h_a. In general, this should be a minimum of 10 feet.

Soil Conditions

The types of soil conditions have a significant influence on the selection of the type of tank foundation. Due to their size and loading, tanks can cause increases in soil stresses to significant depths. Therefore, large settlements may occur in many cases, even if bearing capacity is adequate. Tanks, in general, are capable of withstanding significant uniform settlements without loss of their function.

In general, for good soil conditions, granular soils or medium to stiff clays, gravel ring wall support will be suitable. Exceptions to this would include cases of high uplift due to seismic or pressure loadings requiring anchoring; in these cases, a concrete ring wall should be used.

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In poor soil conditions, soft clays and silts, where anticipated settlements are excessive or potential for a bearing capacity failure exists, surcharging should be considered as an alternate to supporting the tank on piles. Surcharging can be performed by temporary fill. In many cases, the water in the tank itself has been successfully used to surcharge the soils. Settlement measurements must be obtained and rate of filling must be carefully controlled, especially if potential for bearing capacity failure exists. Sand drains are sometimes used to accelerate the process. A 5 to 10 foot thick mat of compacted granular fill is generally provided on top of the soft soils before placing the surcharge.

Pile foundations should be considered as a last resort if the settlement tolerances are very critical or the surcharge is not a feasible option due to time, fill availability, or other constraints.

The soils consultant should always be involved in determining which types of foundations are feasible and which are not. They should also provide a detailed tank hydrotest procedure for monitoring of tank settlement and evaluate the results of settlement data.

Proximity To Slopes

If the tank is located near the top of a slope, as a rough rule of thumb, a minimum setback of twice the height of the slope should be considered. If this setback cannot be provided, or potential for slope instability exists, specific recommendations from the soils consultant should be obtained.

Temperature Fluctuations

Large diurnal temperature fluctuations coupled with large diameter tanks can result in significant movements of the tank bottom plate due to day time thermal expansion and night time contraction. This can result in digging in of the bottom plate into the perimeter foundation. The problem is generally more severe if soil sand is used under the tank perimeter. Foundation and tank should be designed to accommodate this movement at the perimeter of the tank under the shell. A gravel ring wall under the shell instead of sand or a concrete leveling ring would prevent this problem.

Use of Sea Water for Compaction

If sea water is used for compaction of the subgrade soils and the tank foundation, the corrosion potential of the soil in contact with steel should be evaluated. The corrosion potential of the soil should be investigated with the addition of salt water as proposed to be done in the field. As discussed in the corrosion section, contamination with salt is to be avoided in the oil sand or asphalt sand immediately below the tank bottom. Appropriate measures for corrosion protection should be adopted if salt water is to be used. Tank plates should not be placed directly on the contaminated soils.

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DESIGN CONDITIONS

Vertical Loading

Storage tank foundations will be designed to support the tank weight plus the weight of stored product or water, whichever is greater.

Horizontal Loading

Horizontal loads are an important consideration in the foundation design of storage tanks and should always be checked.

Resistance to the overturning moment at the bottom of the shell may be provided by the weight of the tank shell and by the weight of a portion of the tank contents adjacent to the shell for unanchored tanks or by anchorage of the tank shell. For unanchored tanks, the portion of the contents which may be utilized to resist overturning is dependent on the width of the bottom plate under the shell which lifts off the foundation and may be determined as follows:

$$W_L = 7.9 t_b \sqrt{F_{by} \cdot \delta \cdot H_L}$$

Except that W_L will not exceed 1.25 ($\delta\, H_L\, D$).

where

W_L = Maximum weight of tank contents which may be utilized to resist the shell overturning moment, in pounds per foot of shell circumference.

t_b = Thickness of bottom plate under the shell, in inches.

F_{by} = Minimum specified yield strength of bottom plate under the shell, in pounds per square inch.

δ = Design specific gravity of liquid to be stored as specified by the purchaser.

H_L = Height of commodity in tank.

Structural stability of tank is determined by the following equation:

$$\text{Stability Factor, S.F.} = \frac{M}{D^2 (W_t + W_L)}$$

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where

W_t = Weight of tank shell and portion of fixed roof

supported by shell, in pounds per foot of shell circumference.

$$W_t = \frac{W_s + W_r - P}{\pi D}$$

W_S = Total weight of tank shell

W_R = Total weight of tank roof

P = $\rho \dfrac{\pi D^2}{4}$, pressure load in pressurized tank

p = Pressure in tank

D = Diameter of tank, feet.

When stability factor >1.57, tank should be anchored. Refer to API (American Petroleum Institute) 650, Section E.5.1.

When anchors are required for resisting overturning by the design, tension load per bolt will be calculated as follows:

$$t_B = \frac{4M}{dN} - \frac{W}{N} + \frac{P}{N}$$

where

t_B = Design tension load per anchor bolt, in pounds

d = Diameter of anchor bolt circle, in feet

N = Total number of anchor bolts

M = Overturning moment, in foot-pounds

W = $W_S + W_R$; Weight of tank shell and any portion of roof supported by shell in pounds

Anchor bolts will be spaced a maximum of 10 feet apart.

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For the design of anchor bolts, refer to the Structural Engineering Guideline 000.215.1207: Anchor Bolt Design Criteria.

Soil Bearing

In selecting the proper type of foundation, the bearing capacity of the soils is the primary factor. A thorough knowledge of the soil properties is necessary to avoid excessive differential settlement and possible failure.

For the types of soil conditions and their influences on the selection of the type of tank foundations, refer to paragraphs under Soil Conditions.

CONCRETE RING WALL DESIGN

Theory

The thickness of the ring wall is determined from a consideration of bearing pressure. In order to minimize differential settlements, the bearing pressure under the ring wall should be as close as possible to that under the interior portion of the tank. The ring wall depth will be determined from the allowable soil bearing pressures and frost depth requirements, and will be based on soils consultant recommendations.

Soil pressure under interior portion of tank:

$$P = qH + \gamma_s h$$

Soil pressure under ring wall, when the tank shell is at the center of ring wall:

$$P = \frac{1}{t}\left[\omega + \frac{1}{2} tqH + h\, t\, \gamma_c\right]$$

where

P	=	Soil bearing pressure at base of ring wall, (PSF). Must not be greater than the allowable soil bearing, Pa.
q	=	Unit weight of tank liquid, PCF.
H	=	Height of tank liquid, ft.
γ_s	=	Unit weight of backfill inside ring wall, PCF.
γ_c	=	Unit weight of concrete, PCF.

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h	=	Height of ring wall, ft.
t	=	Width of ring wall, ft.
ω	=	Weight of tank per foot of circumference, for tanks with floating roof, or weight of tank shell plus load from tank roof on shell for fixed roof tanks per foot of circumference (LB / FT).
	=	$\left(\frac{W}{\pi D}\right)$

Solving Soil Pressure Equations for t

$$t = \frac{2\omega}{qH - 2h(\gamma_c - \gamma_s)}$$

Minimum thickness, t = 1'- 0"

From a section through the ring wall (refer to Attachment 01), it is seen that the net horizontal force is:

$$F = \frac{1}{2} K_o \gamma_s h^2 + K_o h(qH)$$

or if the terms are rearranged,

$$F = K_o h(qH + \frac{1}{2}\gamma_s h)$$

where

F	=	Radial outward force on ring wall, lb / ft.
K_o	=	At rest earth pressure coefficient.

Note!!! A tank ring wall is a relatively unyielding structure and, therefore, the horizontal earth pressure on the wall will be best represented by the "at rest condition," K_o

$$K_o = (1 - Sin\phi)$$

where

ϕ	=	Angle of internal friction of soil, (degrees)

This horizontal force F causes a hoop tension in the ring wall given by:

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$$T = \frac{FD}{2000}$$

where

T = Axial tension in ring wall, KIPS

D = Tank nominal diameter, FT.

Reinforcing

Because it is assumed that the concrete takes no tensile forces, the wall reinforcing steel must take it all.

$$A_s = \frac{T_u}{\phi f_y} \geq 0.0025ht$$

where

A_s = Required steel area to resist entire axial tension, in.^{2}.

T_u = (1.4) x T (for operating)

T_u = (1.4) x T (for hydrotest)

Note!!! Load factor of 1.4 rather than 1.6 is used because the lateral pressure is caused by the tank, not the soil

ϕ = Strength reduction factor for tension controlled section = 0.90

f_y = Yield strength of reinforcement (KSI)

All circumferential tension steel will be continuous with splices staggered. All splices will be Class B type as given in ACI (American Concrete Institute) 318-02.

Ties will be #4 at 12 inch minimum.

REFERENCES

ACI (American Concrete Institute) 318-02

API (American Petroleum Institute) 650-98

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Structural Engineering
Guideline 000.215.1207: Anchor Bolt Design Criteria

Structural Engineering
Guideline 000.215.1215: Wind Load Calculation

Structural Engineering
Guideline 000.215.1216: Earthquake Engineering

ATTACHMENTS

Attachment 01: (31Mar05)
Forces At Ring Wall Section

Attachment 02: (31Mar05)
Sample Tank Foundation For Anchored Tank

Attachment 03: (31Mar05)
Sample Design 1 - Concrete Ringwall

Attachment 04: (31Mar05)
Sample Design 2 - Anchored Tanks

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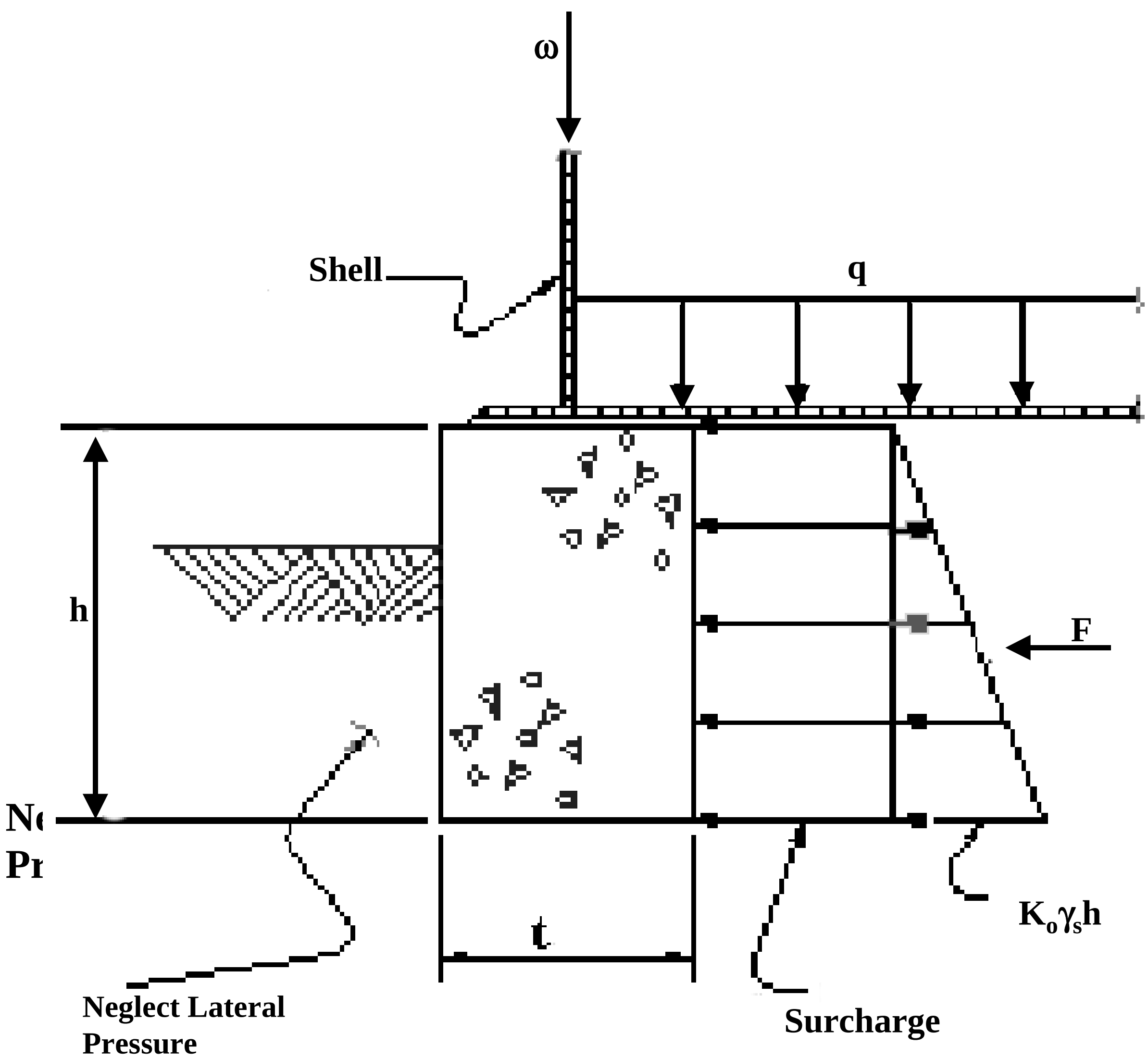
Forces At Plug Wall Section



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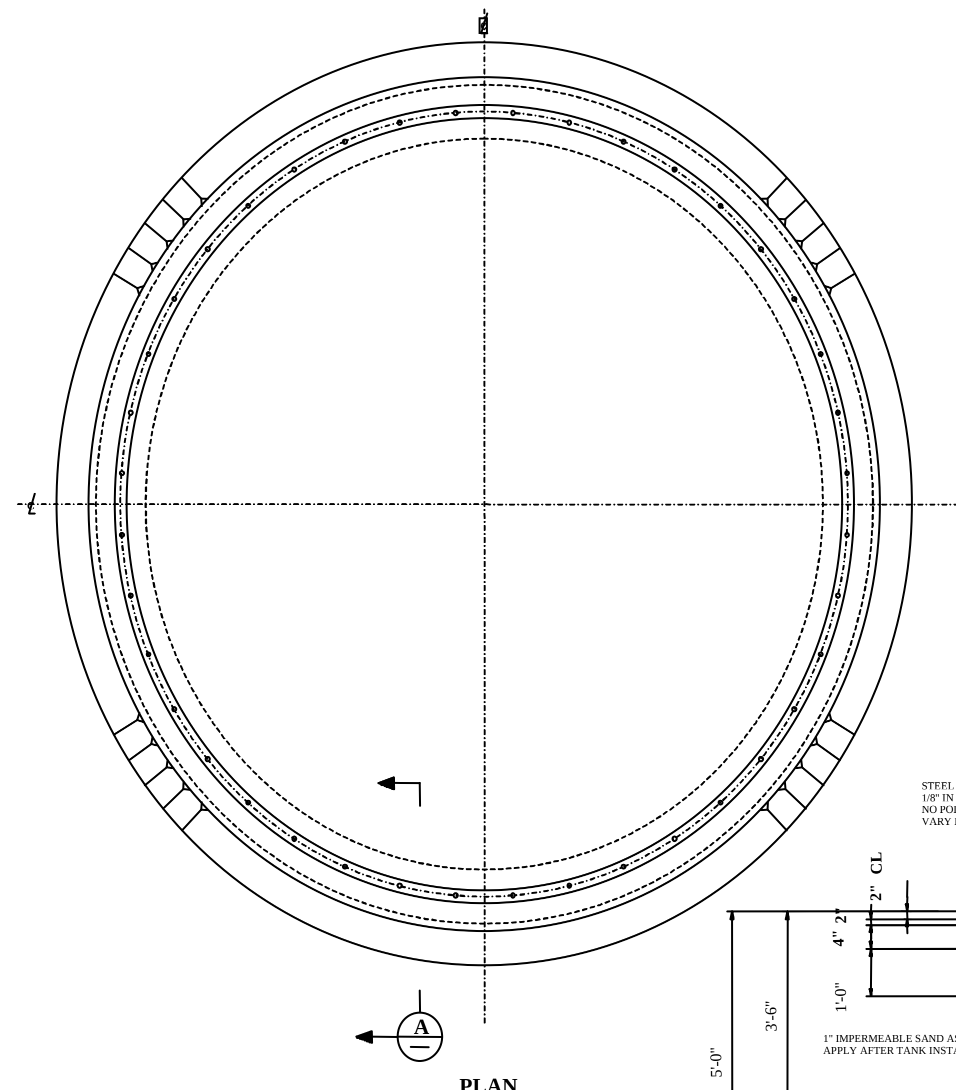
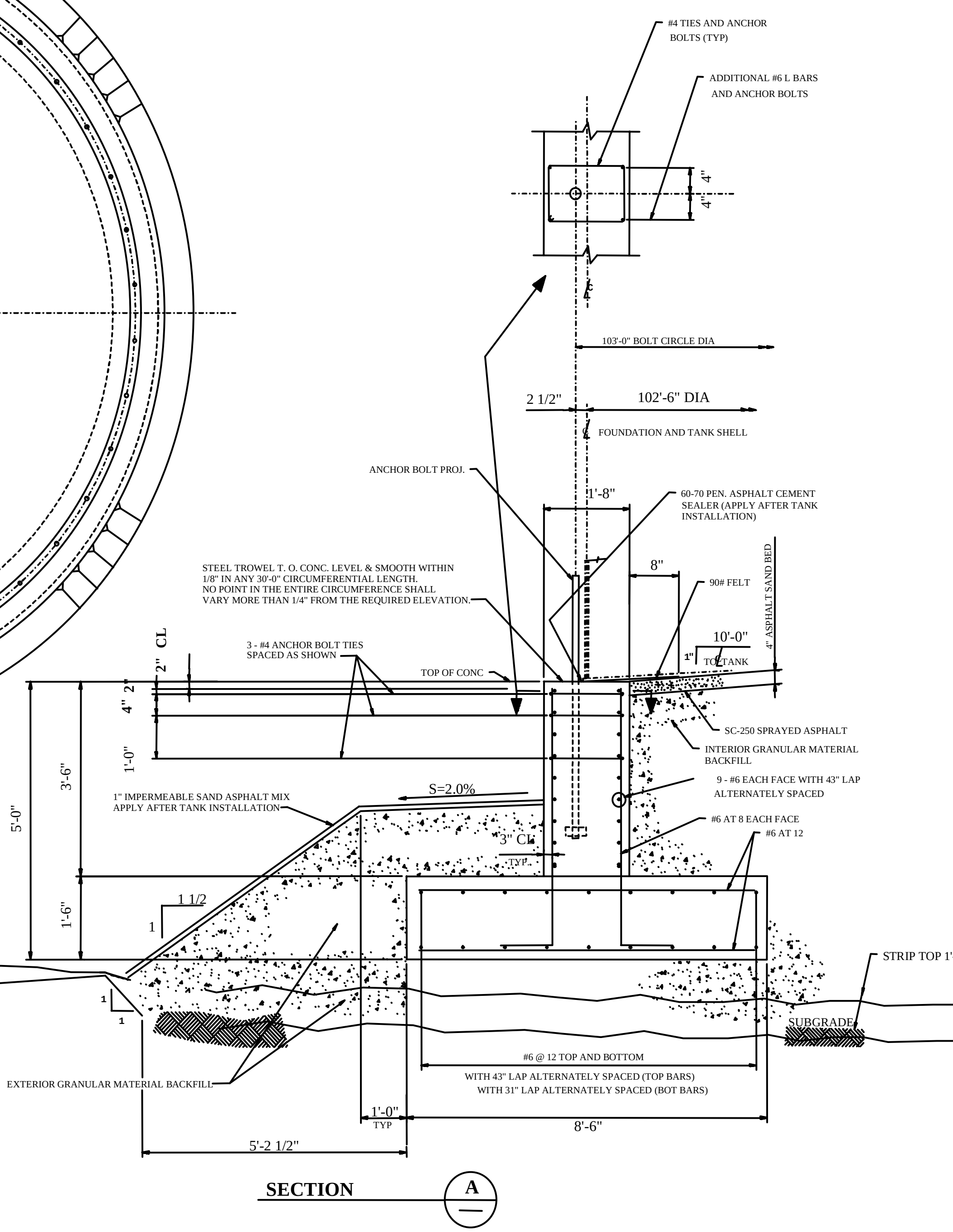
Sample Tank Foundation for Anchored Tank



PLAN



STRIP TOP 1'-0" OF EXISTING GRADE

SECTION A

NOTES:

1. GRANULAR MATERIAL SHALL CONSIST OF CRUSHED ROCK/GRAVEL AGGREGATE MEETING THE REQUIREMENTS OF CALIFORNIA CLASS II AGGREGATE BASE (1 1/2" MAXIMUM SIZE). THE MATERIAL SHALL MEET THE FOLLOWING GRADATION REQUIREMENTS.

SIEVE SIZE	PERCENT PASSING
2"	100
1 1/2"	87-100
3/4"	45-90
NO. 4	20-50
NO. 30	6-29
NO. 200	0-12

2. INTERIOR GRANULAR MATERIAL SURFACE SHALL BE SEALED WITH SC-250 OR APPROVED EQUAL. SPRAYED ASPHALT APPLIED AT THE RATE OF 0.25 GALLON IN ACCORDANCE WITH SQUARE YARD (1.0 LITERS IN ACCORDANCE WITH SQUARE METER). THE SPRAYED ASPHALT SHALL REMAIN UNDISTURBED FOR A PERIOD OF 24 HOURS. THE ASPHALT SHALL BE APPLIED AT TEMPERATURES BETWEEN 40 F AND 8040 F.

3. ASPHALT SAND BED SHALL BE PLACED AS SHOWN OVER THE GRANULAR MATERIAL BACKFILL. SAND TO BE USED SHALL BE FREE OF ALL DELETERIOUS MATERIAL. SALT CONTENT SHALL NOT BE MORE 0.1 PERCENT SAND SHALL BE DRIED TO A FREE MOISTURE CONTENT OF NOT MORE THAN 7 PERCENT BY WEIGHT OF DRY SOIL. THE SAND MUST BE SCREENED THRU 1/4 INCH MESH (MAXIMUM), PRIOR TO ASPHALTING, TO ASSURE ELIMINATION OF ALL CLOTS. CARE SHOULD BE TAKEN IN USING CLEAN MIXING AND HANDLING EQUIPMENT TO ASSURE ASPHALT SAND REMAINS FREE OF THESE CLOTS AND OTHER FOREIGN MATTER. DRY SAND SHALL BE MIXED WITH ABOUT 120 LBS. SC-800 LIQUID ASPHALT TO 1 TON SAND. AFTER PLACING, THE ASPHALT SAND SHALL BE ROLLED OR SHALL BE COMPACTED WITH A COMPACTION PLATE. BEFORE PLACEMENT OF THE TANK, ASPHALT SAND SHALL BE PROTECTED FROM ANY CONTAMINATION (FROM AIR OR SURROUNDING AREA).

4. A SHEET OF 90 LB FELT PAPER SHALL BE PLACED BETWEEN THE TOP OF CONCRETE FOUNDATION AND THE TANK BOTTOM PLATE.

5. EXISTING GRANULAR SURFACE MATERIAL TO BE USED FOR RECOMPACTION MUST BE APPROVED BY THE SOIL ENGINEER.

6. THE TANK SHALL BE FILLED WITH WATER AND THE SETTLEMENT MONITORED AS RECOMMENDED IN ACCORDANCE WITH SOILS REPORT.

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Sample Design 1: Tank Foundations-Concrete Ringwall

Given:

1. f_c = 4000 PSI

   f_y = 60,000 PSI

   γ_c = 150 PCF

2. ϕ = 35 degrees

   γ_s = 100 pcf

   Pa = 4 ksF (Net)

   Soil is contaminated to a depth of 3'- 0" below existing grade elevation ±0.00. Compaction of fill to be 95 percent of maximum dry density at a moisture content close to the optimum (±2 percent) as determined by ASTM D-1557.

   Projected settlement – 2 inches which will be effectively completed by the end of the hydrotest.

3. Corrosive Environment

4. Vendor Data:

   - Total tank empty weight, W = 800 K
   - Tank diameter, D = 90'- 0"
   - Height of tank, H = 48'- 0"
   - Height of tank commodity, H_{OPER} = 41'- 0"
   - Thickness of tank shell and bottom Plate, N = 3/8"
   - Specific gravity of commodity, S = 0.784
   - Weight of steel = 40.8 LB/FT^2-IN
   - Floating roof

5. OTM = 18,660'-K (operating)

6. Elevation at top of ringwall = 1'- 0"

7. ACI 318-02

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Sample Design 1: Tank Foundations-Concrete Ringwall

Design of Ringwall

1. W_t = 40.8 (N)(H)(1.1) = (40.8)(3/8)(48)(1.1) = 808 LB/FT

   (added 10 percent to weight of tank shell for stairs and platforms)

   Note!!! A floating roof is not part of the wall weight.

   Note!!! To determine (OTM) refer to Guideline 000.215.1215 for wind loading and Guideline 000.215.1216 for earthquake loads including sloshing effects.

2. $W_L = 7.9 t_b \sqrt{F_{by} S H_L} = (7.9)(0.375)\sqrt{(36{,}000)(0.784)(41)}$ = 3187 LBS < (1.25)(0.784)(41)(90) = 3616

   OK

   $$SF = \frac{M}{D^2 (W_t + W_L)} = \frac{18660}{90^2 (0.808 + 3.187)} = 0.577 < 1.57$$

   Therefore Anchor bolts are **not** required

   Note!!! Refer to Guideline 000.215.1207 for the design of anchor bolts if required

3. Place top elevation at 1'- 2" to account for settlement due to hydro test. Elevation at bot. of ring wall = -(3'-0") (remove contaminated soil)

   Therefore h = 1.17' + 3.0' = 4.17'

4. γ_s = 115 PCF (95 percent compacted)

   γ_c = 150 PCF

5. Commodity q = (62.4) (0.784) = 48.9 pcf

   Use q_W = 62.4 PCF (any differential settlement will occur during hydrotest)

6. t = 2W / (qWH - 2h(γ_c - γ_s)) = (2 × 808) / [(62.4)(48) - 2(4.17)(150 - 115)] = 0.60'

7. Soil bearing, P = (1 / 1.0)((0.808 + (1/2)(1.0)(0.0489)(48) + (4.17)(1.0)(0.150))

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Sample Design 1: Tank Foundations-Concrete Ringwall

= 2.6 KSF < (4 + (3)(0.115)) GROSS OK

8. a.) $K_o = 1 - \sin\phi = 1 - \sin 35° = 0.43$

b) Test F_T = $K_o h (q_W H + ½ \gamma_s h)$

(H=48'- 0")

= (0.43)(4.17) [(62.4)(48) + ½ (115)(4.17)] = 5.8 K/FT

$F_{u\,(T)}$ = (5.8)(1.4) = 8.12 K/FT CONTROLS

Oper F_O = (0.43)(4.17) ((48.9 × 41) + ½ (115)(4.17))

(H=41'- 0")

= 4.02 K/FT

$F_{u\,(o)}$ = (4.02)(1.4) = 5.63 K/FT

c) T_u = FuD / 2000 = (8120)(90) / 2000 = 365.4 k

d) A_s = $T_u / \phi f_y$ = 365.4 / (0.9)(60) = 6.77 in2

$A_{s(min)}$ = (0.0025)(12) ((4.17)(12)) = 1.5 in^2 < 6.77 in^2 Refer API 650- 98 Section B.4.2.3.d

e)

Use 6 #7's each face, $A_{S(TOT)}$= 7.2 in^2 with class “B” lap splices alternately spaced and min.#4 ties at 12 inches

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Sample Design 2: Tank Foundations - Anchored Tanks

Given:

1.) f_c = 4000 PSI

f_y = 60,000 PSI

g_c = 150 PCF

2.) ϕ = 35 degrees

γ_s = 115 pcf

P_a = 4 ksf (net)

3.) Vendor Data:

- Total Wt of tank shell, W_s = 203,346 LBS
- Total Wt of tank roof, W_R = 121,640 LBS (cone roof)
- Total Wt of tank content, W_T = 21,497,040 LBS
- Tank diameter, D = 102'- 6"
- Tank Height, H = 48'- 0"

4.) Total overturning Moment applied to bottom of tank due to seismic, M = 101,023,107 lb-ft

Uplift Force due to overturning moment, U_s = 12,242 LB/FT

(Refer to Practice 000.215.1216 for earthquake loads, including sloshing effects, and Practice 000.215.1207 for the design of anchor bolts.)

Design of Foundation:

Stability Factor = $M / D^2 (W_t + W_L)$

$W_t = (W_s + W_R) / \pi D = (203{,}346 + 121{,}640) / ((\pi)(102.5)) = 1009$ LB/FT

Weight of tank contents which may be utilized to resist the shell overturning moment is given by:

$$W_L = 7.9 t_b \sqrt{F_{by} S H_L}$$

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Sample Design 2: Tank Foundations - Anchored Tanks

Where:

- Thickness of bottom plate under the shell, t_b = 5/16 "
- Yield strength of bottom plate Fby = 36,000 KSI
- Specific Gravity of commodity S = 0.784
- Height of commodity in tank H_L = 41'- 0"

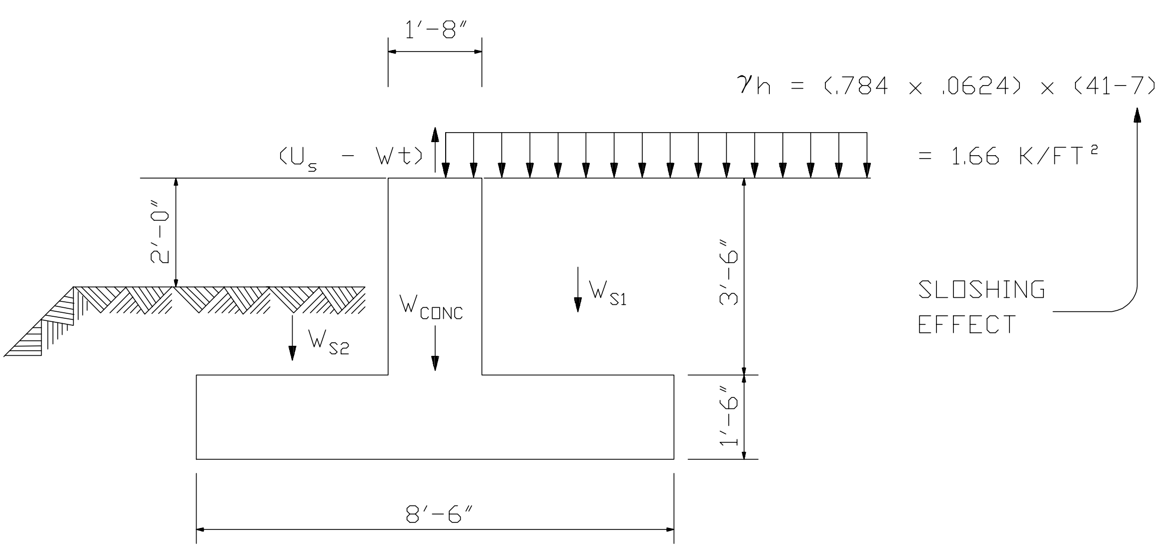
Therefore $W_L = (7.9)(\frac{5}{16})\sqrt{(36{,}000)(0.784)(41)}$

=2656 LB/FT of circumference < (1.25)(0.784)(41)(102.5) = 4118 OK

Therefore Stability Factor = 101,023,107 / $(102.5)^2$ (1009 + 2656) = 2.62 > 1.57

Therefore The tank is structurally unstable and anchoring the tank is required.

a.) Condition No. 1: Maximum Uplift



Total Uplift force due to overturning: $(U_S - W_t)$ = 12.242 - 1.009 = 11.233 K/FT

$W_{CONC.}$ = ((20/2)(3.5) + (8.5)(1.5)) (0.15) = 2.79 K/FT

W_{S1} = ((8.5 - 1.67) / 2) (3.5)(0.115) = 1.37 K/FT

TANK FOUNDATIONS

Sample Design 2: Tank Foundations - Anchored Tanks

W_{S2} = ((8.5 - 1.67) / 2) (1.5)(0.115) = 0.59 K/FT

Wt of tank content = (1.66)(8.5 / 2) = 7.06 K/FT

Total Downward Force = 2.79 + 1.37 + 0.59 + 7.06 = 11.81 K/FT > 11.233 K/FT **OK**

FOOTING TOP REBARS SHALL BE DESIGNED FOR CONDITION NO. 1

b.) Condition No. 2: Maximum Compression

Compression Force due to OTM, U_S = 12.242 K/FT

W_t = 1.009 K/FT

Wt of tank content = (0.784)(0.0624) (41 + 7) (8.5 / 2) = 9.98 K/FT

↑ *Sloshing effect*

Wt of Concrete and Soil = 2.79 + 1.37 + 0.59 = 4.75 K/FT

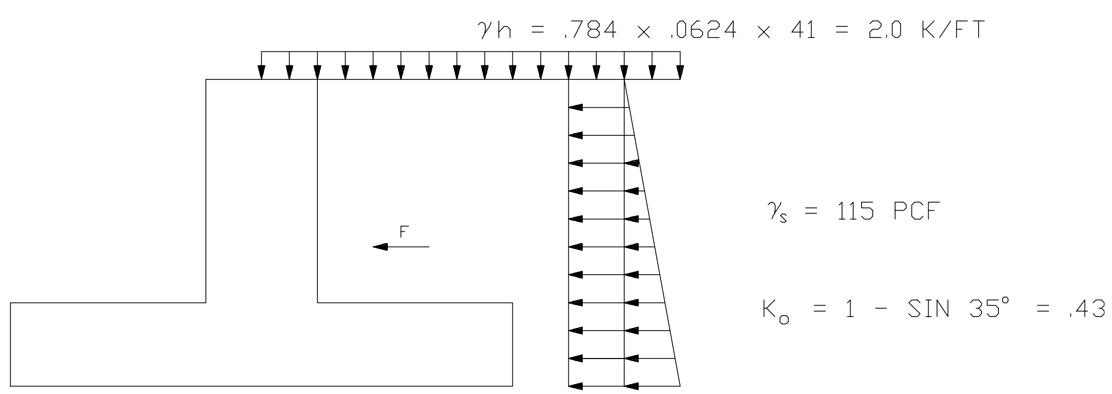
Max. Compressive load = 12,242 + 1.009 + 4.75 + 9.98 = 28.0 K/FT

Soil Bearing Pressure = 28.0 / (8.5 × 1) = 3.29 KSF (Gross)

= 3.29 - (3 × 0.115) = 2.95 (Net) < 1.33 × 4.0 **OK**

PEDESTAL AND FOOTING BOTTOM REBARS SHALL BE DESIGNED FOR CONDITION NO. 2

c.) Condition No. 3: Maximum Static Load



FLUOR

TANK FOUNDATIONS

Sample Design 2: Tank Foundations - Anchored Tanks

(Oper)

F_o = $K_o\, h\ (qH + ½\ \gamma_s h)$

= (0.43)(3.5) ((0.784)(0.0624)(4) + ½ (0.115)(3.5))

= 3.32 k/ft

$F_{U\,(O)}$ = (3.32)(1.4) = 4.65 K/FT

(Test)

F_T = (0.43)(3.5) ((0.0624)(48) + ½ (0.115)(3.5))

= 4.81 K/FT

$F_{U\,(T)}$ = (4.81)(1.40) = 6.734 K/FT **←CONTROLS**

Therefore

T_U = $F_U D$ / 2000 = (6734)(102.5) / (2000) = 345 K

A_S = $T_U / \phi f_y$ = 345 / (0.9)(60) = 6.4 in^2

$A_{S(MIN)}$ = (0.0025)(20)(3.5)(12) = 2.1 in^2 < 6.4 in^2

Use	18 - #6, A_s = (18)(0.44) = 7.92 in^2
	9 - #6 Each face with Class “B”
	Lap Splice alternately spaced